

PORTONIAN PUSH

HYBRID INFRASTRUCTURE-ASSISTED MANET FOR INDUSTRIAL EMERGENCY RESILIENCE

F20MX Mobile Communications and Programming (GA)

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MANET Technical Background

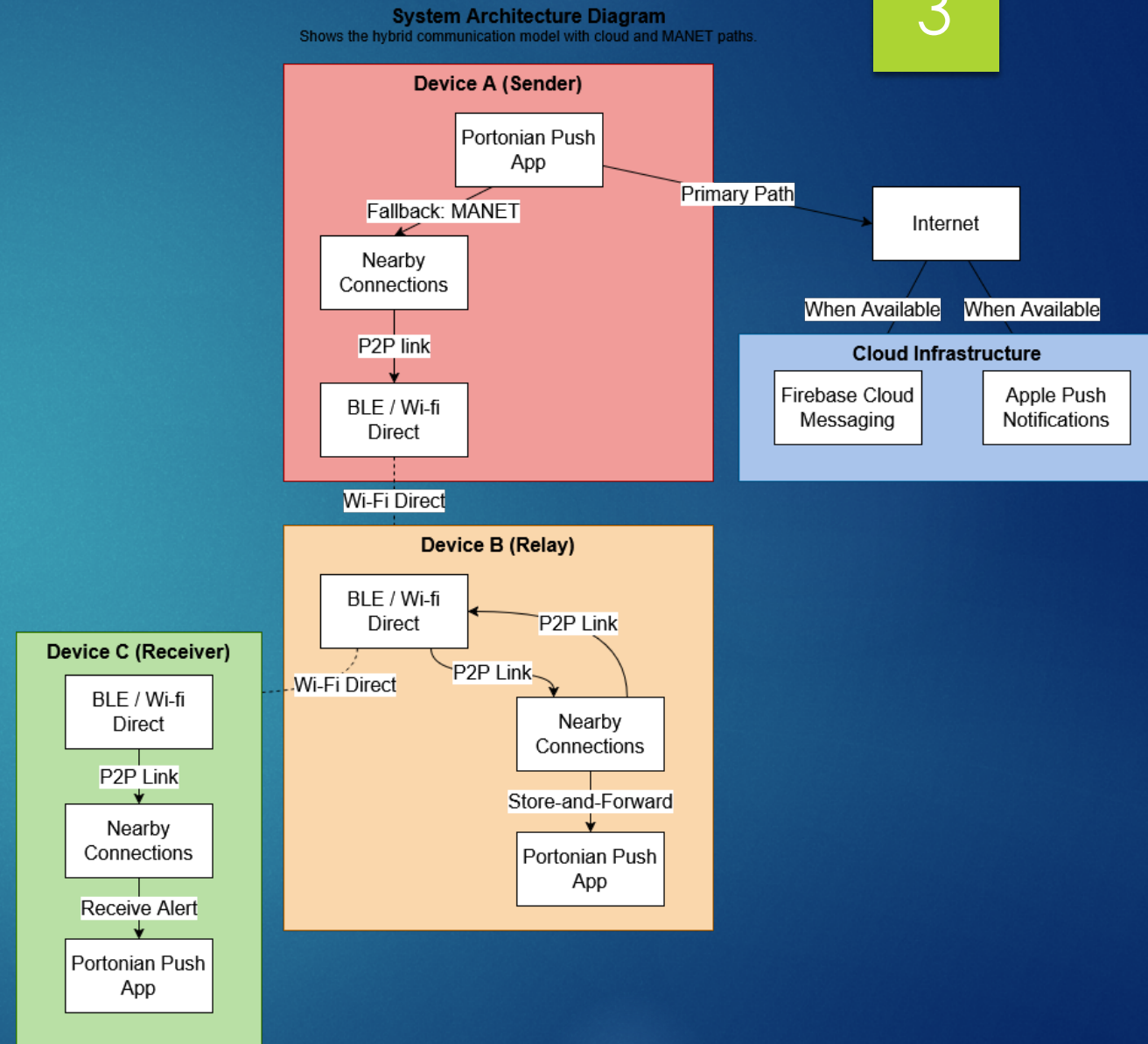
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- ▶ **MANET:** Mobile Ad-Hoc Network - no fixed infrastructure
- ▶ Devices act as endpoints *and* relays
- ▶ **Multi-hop:** messages propagate through intermediate nodes
- ▶ **Trade-off:** Resilience vs. range and complexity
- ▶ **Relevance:** Reliability when infrastructure fails

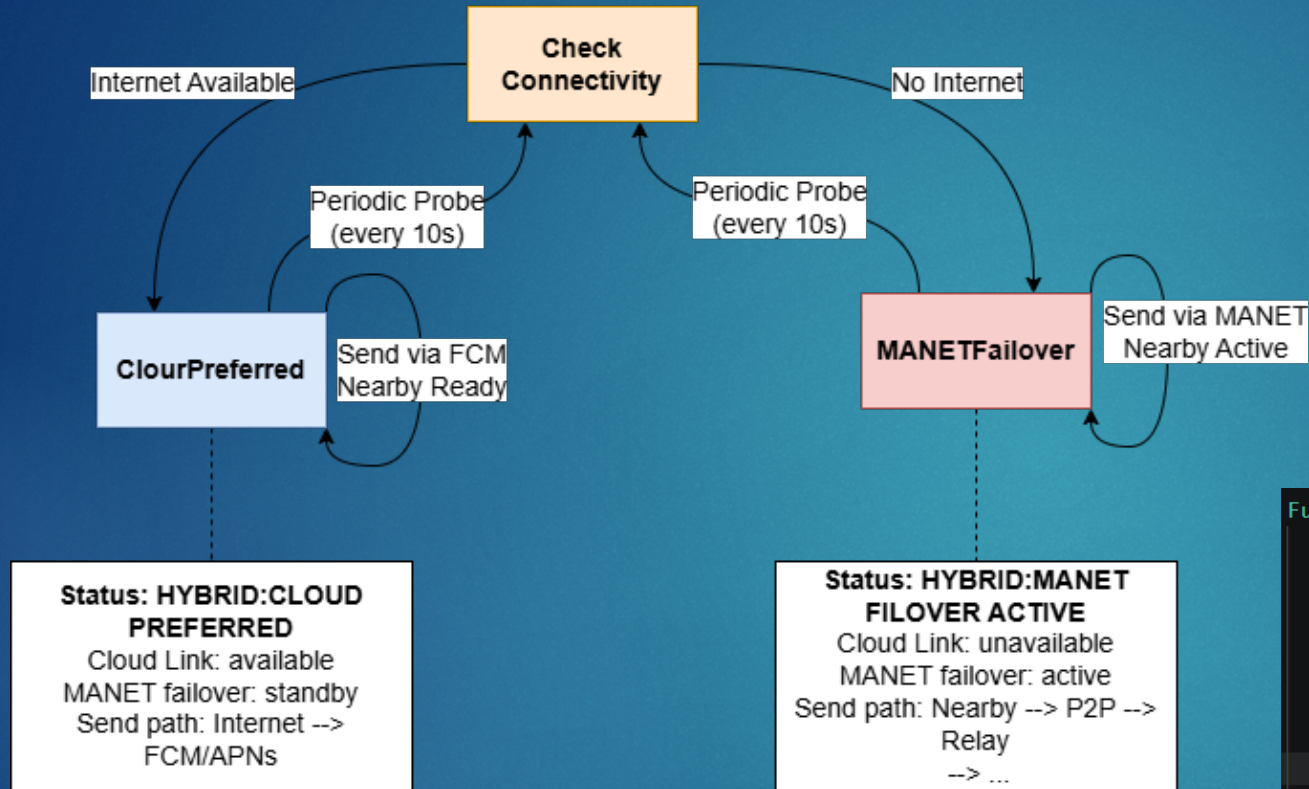


Hybrid Architecture

- ▶ **Primary:** Cloud (Internet available)
 - FCM, APNs—fast, reliable
- ▶ **Fallback:** MANET (Internet unavailable)
 - Nearby Connections, Wi-Fi Direct, Bluetooth
- ▶ **Automatic failover:** Probe every 10 seconds
- ▶ Seamless mode switching



Hybrid Failover State Machine



```

Future<bool> probeInternetReachability() async {
  try {
    final List<InternetAddress> lookupResult = await InternetAddress.lookup(
      "example.com",
    ).timeout(const Duration(seconds: 3));
    return lookupResult.isNotEmpty &&
      lookupResult.first.rawAddress.isNotEmpty;
  } catch (_) {
    return false;
  }
}

```


Applications & Use Cases

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Industrial Incident Scenarios:

- ▶ **Industrial incidents:** Chemical spill, infrastructure failure
- ▶ Mobile networks become congested or unavailable
- ▶ **First responders:** Need alerts without infrastructure
- ▶ **Real-world:** Military tactical mesh, emergency responder networks
- ▶ **Scenario:** Alert relays worker-to-worker across site



Advantages

- ▶ **Resilience:** No single point of failure; works when infrastructure is down
- ▶ **Hybrid flexibility:** Cloud when available, MANET when needed
- ▶ **Low cost:** Uses existing devices; software-only deployment
- ▶ **Measurable:** Can instrument for handshake, latency, throughput, PDR
- ▶ **Redundancy:** Multiple relay paths increase reliability

Limitations and Future Work



Reliability: First message ~100%; repeated sends inconsistent



Handshake overhead: 9–11.5 seconds to connect



Range: Wi-Fi Direct ~100m, Bluetooth ~10–50m



Background: Only shows alerts when app is foregrounded



Security: No digital signatures or trust model (PoC)



OS constraints: Android may re-enable Wi-Fi; background limits

Trends & Future Developments

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5G-assisted MANET: Faster failover, edge computing

Advanced routing: Machine learning for optimal relay selection

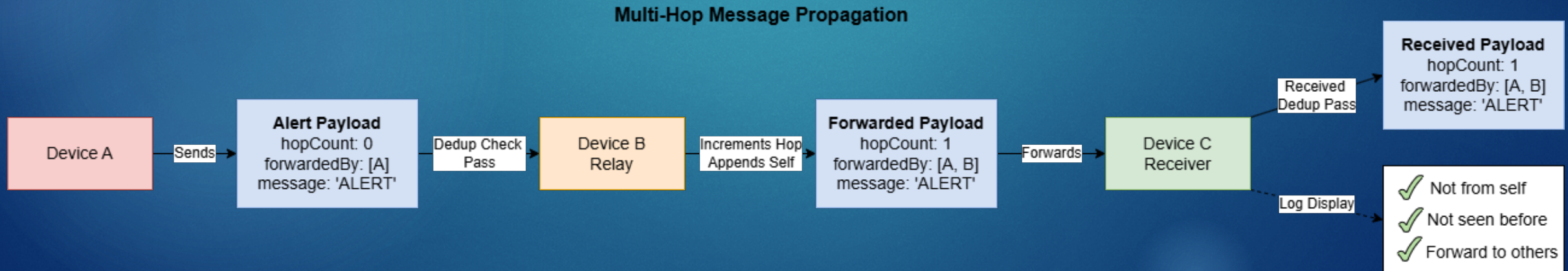
Blockchain: Distributed authentication for decentralised trust

Wi-Fi HaLow: 900 MHz standard, 1+ km range

Dual-radio: Separate control and data channels

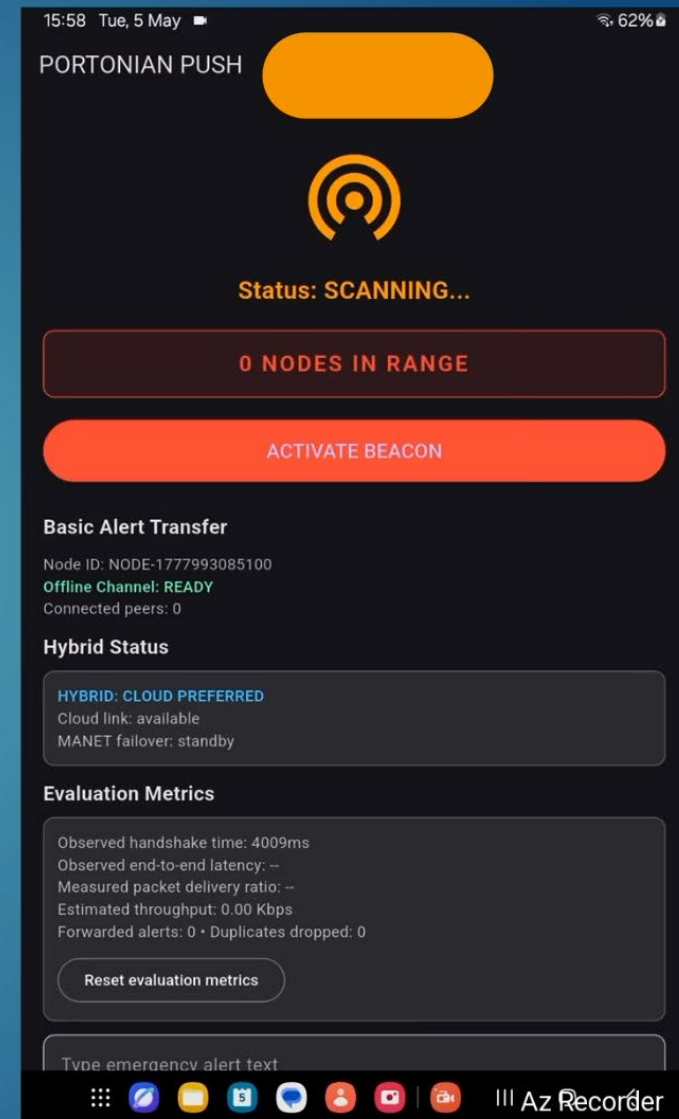
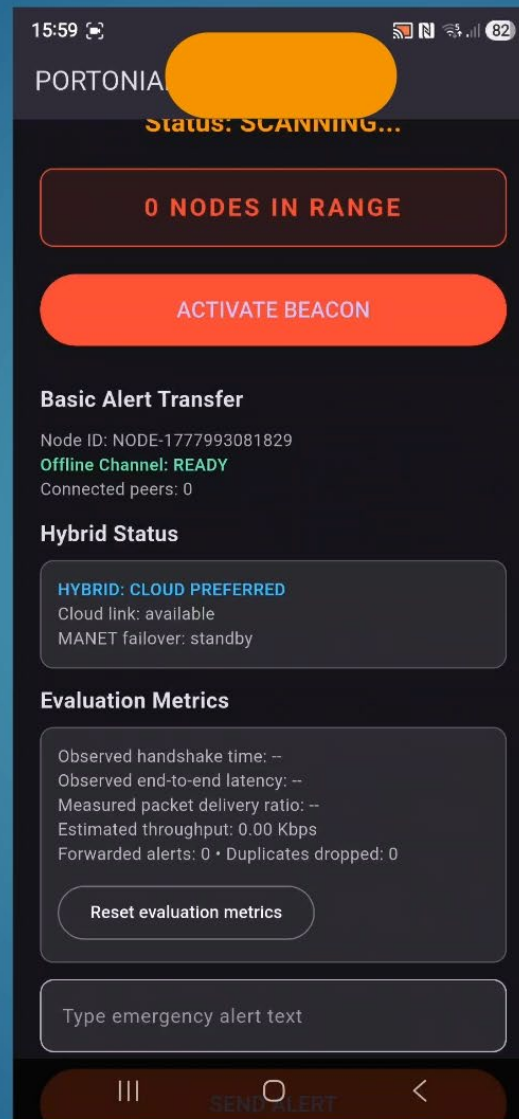
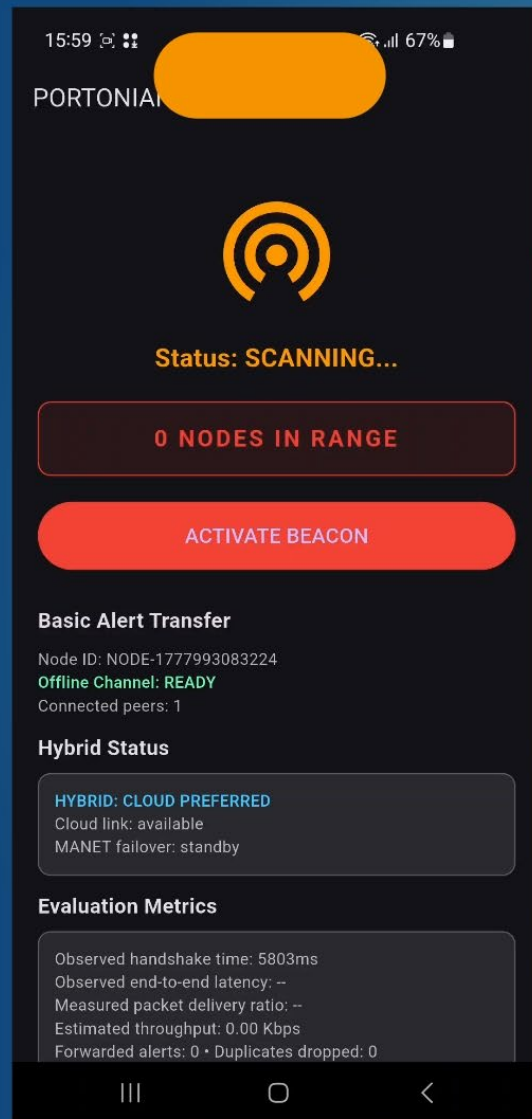
Main Features

- ▶ Nearby device discovery
- ▶ Alert sending and receiving
- ▶ Multi-hop forwarding
- ▶ Hybrid connectivity status
- ▶ Logging and evaluation metrics



Demo

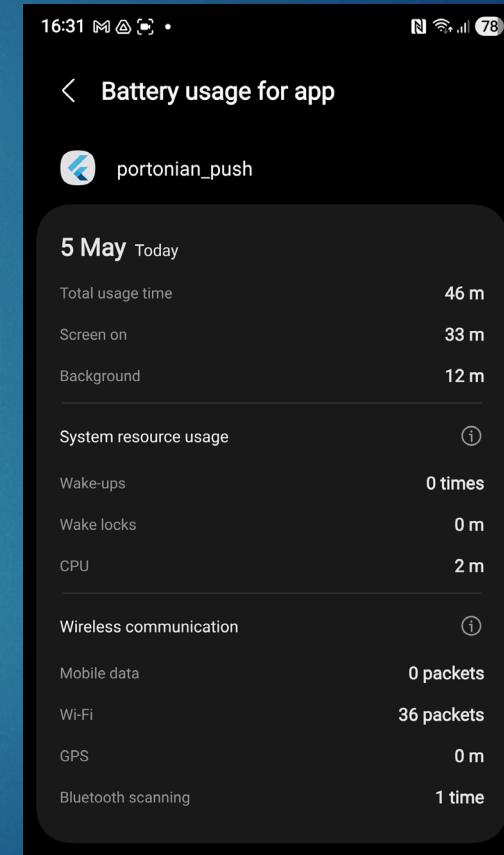
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Measured Performance

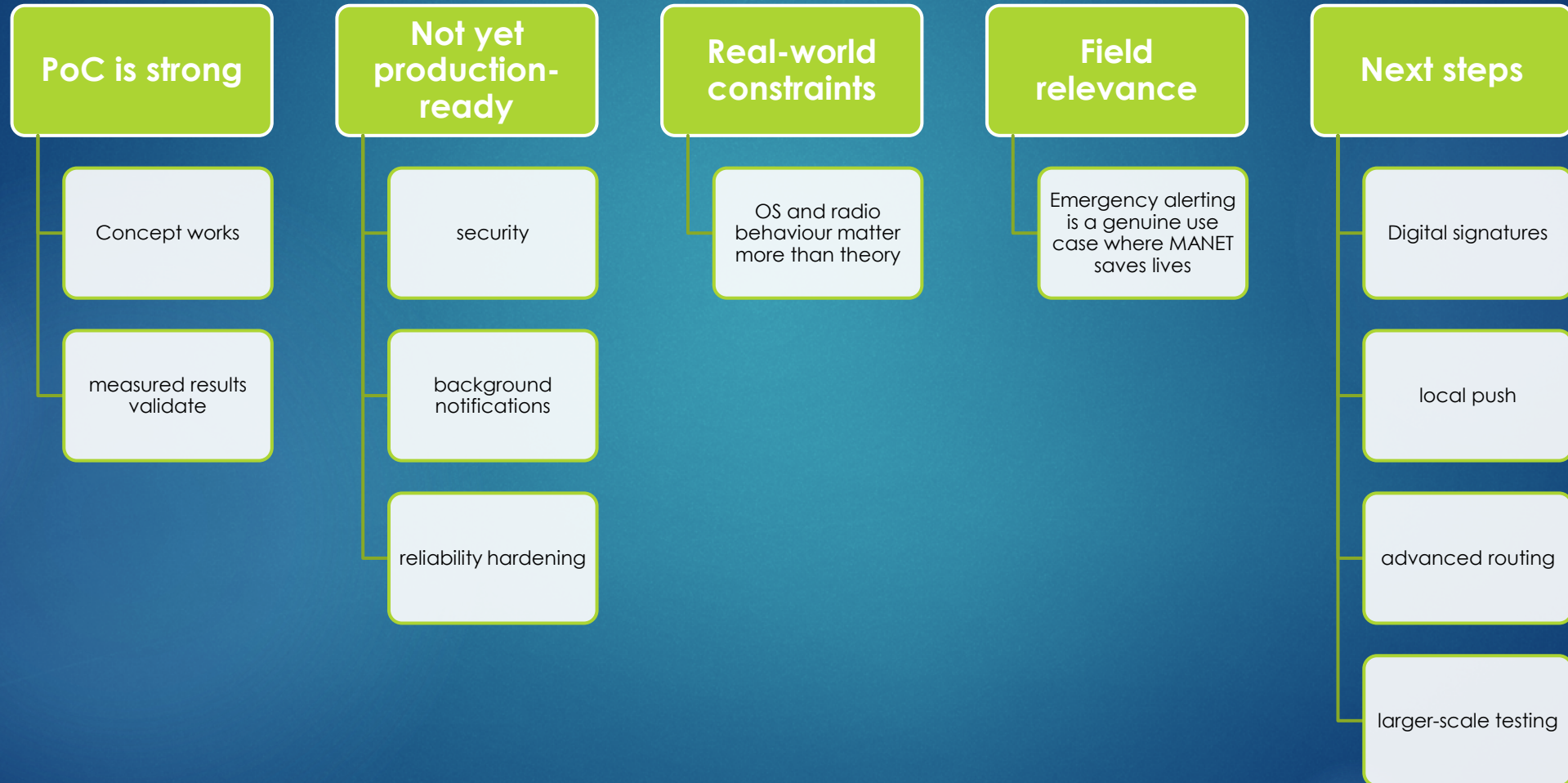
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Metric	Device A Result	Device B Result	Device C Result
Handshake	26720ms	14093ms	37195ms
Latency	-1075ms	1712ms	932ms
Throughput	0.02Kbps	0.03kbps	0.02kbps
Packet Delivery Ratio	100%(1/1)	100%(1/1)	--
Network Stability	★★★★☆☆	★★★★☆☆	★★★☆☆☆



Conclusion

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Thank You!

Repository: https://github.com/SusBin/portonian_push/

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